

Active Suspension Control for Quarter Car Model

Group 10:

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Abstract—Suspension systems are designed to enhance vehicle comfort, safety, and performance. Passive suspension systems are limited by their inability to adapt to varying driving conditions, leading to compromises between comfort and handling. Active suspension systems, however, can adjust in real time for optimal performance. This report investigates the performance of two advanced control strategies for active suspension in a quarter-car model: H-infinity ($\mathcal{H}_\infty$) and Model Predictive Control (MPC). The performance of the $\mathcal{H}_\infty$ controller, which aims to minimize the worst amplification of road disturbances, is tested in three different modes: comfort, balanced and performance. The robustness of the $\mathcal{H}_\infty$ controller is further enhanced using a μ -synthesis controller. The $\mathcal{H}_\infty$ controlled system shows instability, but can be stabilized by μ -synthesis. The performance of a Model Predictive Control (MPC) approach, which optimizes control action based on predictions of future system behavior, was investigated and compared to Linear Quadratic Regulator (LQR) control. These strategies and their modes were compared based on tire force for road holding, and suspension deflection and body acceleration for comfort. The results show that the investigated control strategies offer significant improvements compared to passive systems, each having unique advantages under different conditions.

I. INTRODUCTION

Road disturbances, caused by uneven road surfaces, bumps, and potholes, can create unwanted oscillations resulting in an uncomfortable rides and decreased vehicle handling and stability. The automotive research into comfort, safety, and performance is everlasting. Here, active suspension systems could play a significant role due to their ability to respond dynamically to changing road conditions, unlike passive suspension systems [1]. In this report, two advanced control strategies for active suspensions are considered: H-infinity ($\mathcal{H}_\infty$) and Model Predictive Control (MPC). The primary objective is to investigate the performance of these strategies in an active suspension system for a quarter-car model compared to passive suspension. Here, good performance is measured by minimal fluctuations in tire force for proper road holding and minimal low-frequency body acceleration for comfort.

First, a $\mathcal{H}_\infty$ -controller is implemented and its performance is analysed. This robust control technique minimizes the largest amplification of the disturbances. Three modes were designed into the $\mathcal{H}_\infty$ -controller: comfort, balanced, and performance, as these are today's most commonly used driving modes. The modes contain optimised response characteristics for the corresponding objectives. The $\mathcal{H}_\infty$ -controller is developed around a nominal plant model. The three modes are then simulated and compared. To further robustify the controller, uncertainties and model errors are introduced and a μ -synthesis controller is designed. μ -synthesis can be seen as an extension of $\mathcal{H}_\infty$. It still solves the same problem, but minimises the worst-case gain across the entire uncertainty space. Finally, the stabilization of the μ -synthesis controller is analyzed.

The second control technique investigated is a model predictive control (MPC) approach. This controller predicts the system's future behaviour and solves an optimisation problem each timestep to find the optimal control action. Weight factors can be tuned, reflecting individual states and control actions, and constraints can be set to ensure the limits of the system and actuator are sustained. To investigate the performance of the MPC controller, the performance of a Linear-Quadratic Regulator (LQR) controller is simulated and used as a reference for the MPC design.

In this report, an explanation of the quarter car model and its differential equations are provided in Section II. Next, the design and results of the $\mathcal{H}_\infty$ -controller and μ -synthesis are discussed in Section III. This is followed by an explanation and evaluation of the MPC controller and its results in Section IV. Lastly, a conclusion is provided in Section V, discussing the results and possible future work.

II. QUARTER CAR MODEL

A quarter car model is a simplified model of a vehicle's suspension system focusing on a single wheel and its components. By isolating a quarter of a car, the model can capture the essential dynamics of the suspension system while reducing computational complexity. A diagram of the quarter car model is shown in Figure 1. The model parameters and their values can be found in Table I. Here, the sprung mass (m_s) represents the vehicle body and z_s its vertical position. The unsprung mass (m_u) represents the wheel and axle assembly with vertical position z_u , and the spring (k_s) and damper element (c_s) represent the suspension. For a quarter car with active suspension, there is an additional 'slow active' actuator. The force this actuator applies on the system can be controlled and will consequentially be control input u . The tire absorbs road irregularities and provides contact with the road by flexing and is treated as a spring with stiffness k_t . The sprung mass natural frequency $\omega_{n,s}$ is 7.48 rad/s, which can be seen as the frequency that amplifies bouncing the most.

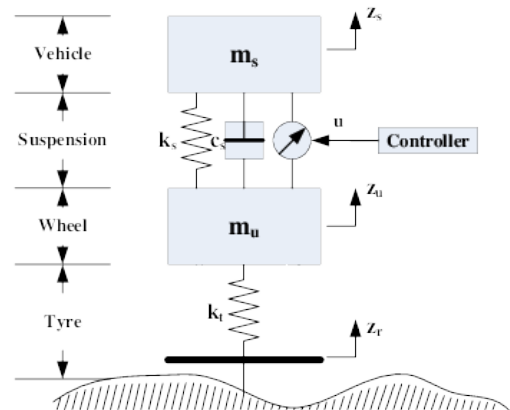


Figure 1: Quarter Car Model [2]

Parameter	Symbol	Value	Unit
Sprung mass	m_s	410	kg
Unsprung mass	m_u	45	kg
Tire vert. stiffness	k_t	230,000	N/m
Suspension vert. stiffness	k_s	25,500	N/m
Damping ratio	c_s	1840.7	Ns/m

Table I: Quarter car model parameters

Differential equations that describe the motion of the sprung and unsprung masses in response to road disturbances and the control input are derived from the model. These equations are shown in Equation 1 and Equation 2 respectively.

$$m_s \ddot{z}_s + c_s(\dot{z}_s - \dot{z}_u) + k_s(z_s - z_u) = -u \quad (1)$$

$$m_u \ddot{z}_u + c_s(\dot{z}_u - \dot{z}_s) + k_s(z_u - z_s) + k_t(z_u - z_r) = u \quad (2)$$

Based on these equations, the following dynamic system for states x can be obtained:

$$x = \begin{bmatrix} z_u - z_r \\ \dot{z}_u \\ z_s - z_u \\ \dot{z}_s \end{bmatrix}, \quad \dot{x} = Ax + Bu + Gz_r \quad (3)$$

$$y = Cx + D_u u + D_w z_r$$

where G is the road disturbance matrix, D_w is the output disturbance matrix. The matrices describing the system are shown in Figure 4. The output of the system are vehicle body acceleration, tire force and travel suspension, and the input is the hydraulic force from the actuator.

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -\frac{k_t}{m_u} & -\frac{c_s}{m_u} & \frac{k_s}{m_u} & \frac{c_s}{m_u} \\ 0 & -1 & 0 & 1 \\ 0 & \frac{c_s}{m_s} & -\frac{k_s}{m_s} & -\frac{c_s}{m_s} \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ \frac{m_s}{m_u} \\ 0 \\ -1 \end{bmatrix},$$

$$G = \begin{bmatrix} 0 \\ \frac{k_t}{m_u} \\ 0 \\ 0 \end{bmatrix}, \quad C = \begin{bmatrix} k_t & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & \frac{c_s}{m_s} & -\frac{k_s}{m_s} & -\frac{c_s}{m_s} \end{bmatrix}, \quad (4)$$

$$D_u = \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix}, \quad D_w = \begin{bmatrix} -k_t \\ 0 \\ 0 \end{bmatrix}$$

III. $\mathcal{H}_\infty$ CONTROL

The block diagram of the $\mathcal{H}_\infty$ controller, K , connected to the generalized plant, P , is shown in Figure 2. The generalized plant includes the dynamics of the quarter car model as well as the uncertainty and performance dynamics of the system. External inputs, w , can come from disturbances, noise or reference signals, while z contains the error signals generated by applying weights to the outputs and the output disturbances. The optimisation algorithm penalizes the gain of the system to these error signals, since the inverse of the weights define an upper bound for the sensitivity transfer function. Thus, the $\mathcal{H}_\infty$ controller can be defined as an optimal control technique that aims to counteract the effect of w on z by finding a controller that minimizes the maximum singular value of the sensitivity transfer functions [3]. As a

result, the magnitude bode plot of the sensitivity functions are pushed below the closed loop targets. For suspension control, the designed controller minimizes the amplification of the road disturbance while using the least amount of actuator force possible in the process.

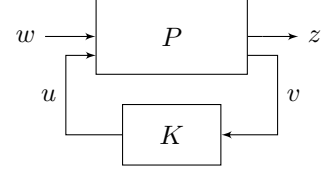


Figure 2: Block diagram of the H-infinity controller

The $\mathcal{H}_\infty$ controller is designed around a nominal model of the system. To implement the uncertainties and model errors of the actuator, μ -synthesis can be used to find the worst case solution for a controller to ensure robustness. The uncertainty is introduced to the block diagram as shown in Figure 3.

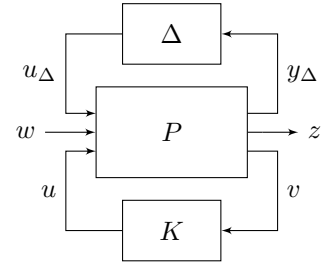
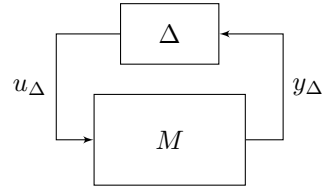


Figure 3: Block diagram with uncertainties

Since there is no direct method to find the $\mathcal{H}_\infty$ norm (μ) of the uncertain system, the block diagram is further simplified to the $M - \Delta$ structure as shown in Figure 4.

Figure 4: $M\Delta$ -structure for robust stability analysis

Scaling matrices D and D^{-1} can be introduced to the entrances and exits of both M and Δ , resulting in the lower path becoming $DM D^{-1}$ and, similarly, the upper path $D\Delta D^{-1}$. Using this final structure, the uncertainties can be minimised by alternating between optimising D and the controller K within M . This method is called D-K synthesis and is defined by the following optimisation problem, resulting in the μ -synthesis controller:

$$\min_K \left(\min_{D \in \mathcal{D}} \|DM(K)D^{-1}\|_\infty \right) \quad (5)$$

The above-mentioned controller designs, $\mathcal{H}_\infty$ and μ -synthesis, are tested on the active suspension quarter car model. The MathWorks code written and explained by Brian Douglas is used [4]. The code is edited to include the

plant defined in II. **Quarter Car Model** including an actuator modified to account for the difference in model parameters. Initially, an $\mathcal{H}_\infty$ controller is implemented on this plant by using MATLAB's `hinfsyn(.)` function. Next, the effects of an uncertain actuator model is analysed, and a robust μ -synthesis controller is designed by MATLAB's `musyn(.)` function.

A. $\mathcal{H}_\infty$ Results

Three different controllers are designed for comfort, handling and balanced driving modes that display different suspension actions. These modes are set by optimising the system with weights W_{ab} and W_{sd} for the performance signals of body acceleration and suspension deflection, respectively. Low body acceleration and high suspension deflection are accepted for comfort mode by optimising the system with $0.99W_{ab}$ and $0.01W_{sd}$; $0.5W_{ab}$ and $0.5W_{sd}$ are set for the balanced mode; finally, for the handling mode $0.01W_{ab}$ and $0.99W_{sd}$ are used.

Bode magnitude plots of the closed loop transfer functions that relate the outputs to the road disturbance are presented in Figure 5. The suspension deflection for the comfort mode has less magnitude reduction than the other two modes as expected. Similarly, the handling mode displays a larger body acceleration magnitude at higher operating frequencies.

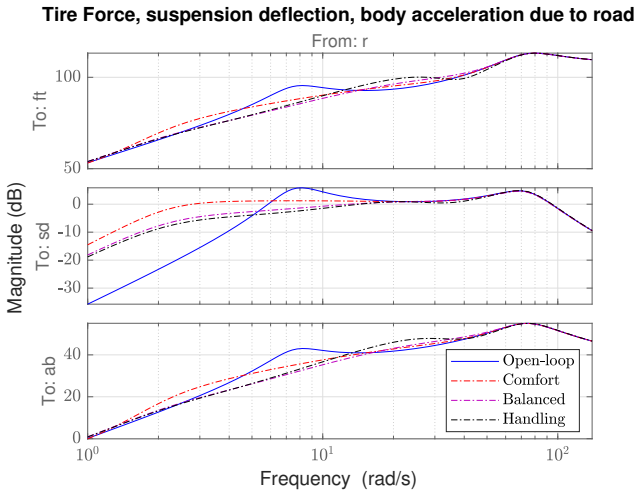


Figure 5: Bode Magnitude Plots for the Designed Comfort, Balanced and Handling Mode Systems

The outputs of the plant and the control input of the closed loop systems are plotted in Figure 6. It can be seen that the settling time for the disturbance rejection is larger for the open loop system compared to the closed loop responses. Thus, the effect of the road bump is felt by the user for a shorter period of time when the suspension system is under the influence of any of the controllers. The handling mode, which is expected to provide the driver with more feedback, shows higher amplitude oscillations for the body acceleration response, while displaying a lower settling time for the suspension deflection. Forcing the suspension deflection to zero immediately after the bump requires a higher control

input to be applied to the plant, thus the handling mode presents with a larger control input, while the comfort mode provides the least. As a result, the user will be able to sense the road profile better in the handling mode, while experiencing smoother transitions through the disturbances in comfort mode.

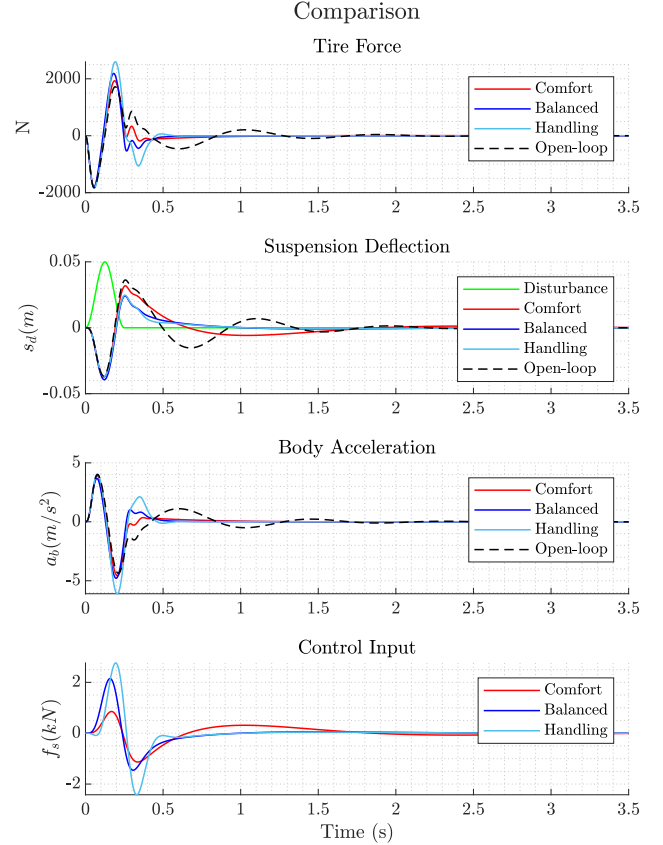


Figure 6: Outputs and Control Input for the Controllers

B. μ -synthesis Results

μ -synthesis tries to minimise the worst-case gain across the entire uncertainty space rather than focusing on the nominal plant only. As a result, it is expected to provide a worse response profile than the $\mathcal{H}_\infty$ controller. To be able to compare the two controllers, the same weights on the error signals are chosen. Therefore, the comfort mode of $\mathcal{H}_\infty$ is selected and the μ -synthesis is run with the same weights of $0.99W_{ab}$ and $0.01W_{sd}$ on the outputs. The D-K iterations resulted in a structured singular value of 0.965. Since this value is less than 1, it indicates that robust performance has been achieved and the system can continue its operation even under perturbations and disturbances.

The controller is tested on the nominal plant and the responses are plotted on Figure 7. As expected, the tire force and the body acceleration for the μ -synthesis controller takes longer to reject the disturbance, while it shows less magnitude reduction between the consecutive peaks of the responses compared to the $\mathcal{H}_\infty$ controller. The suspension deflection also shows a slightly slower approach to zero for μ -synthesis. Finally, the robust μ -synthesis controller requires higher control input to provide the observed responses.

Comfort Mode Comparison

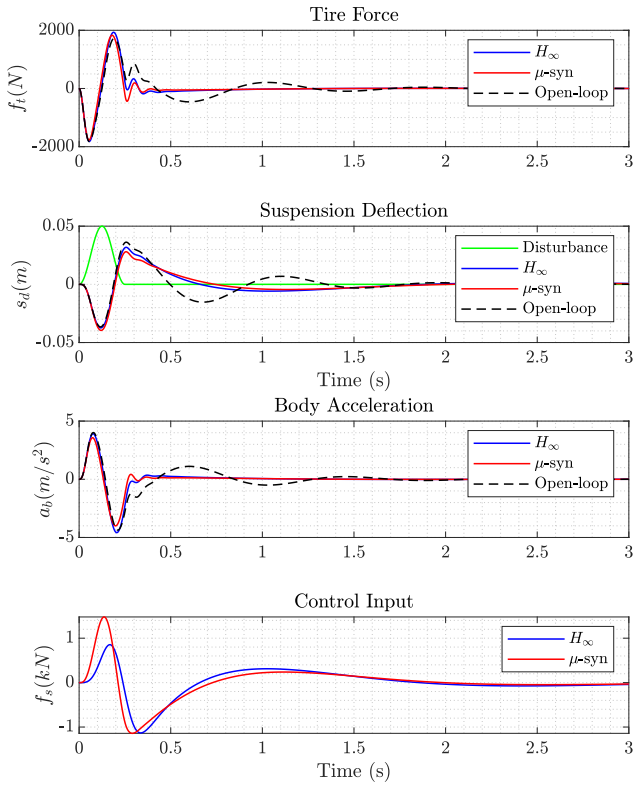
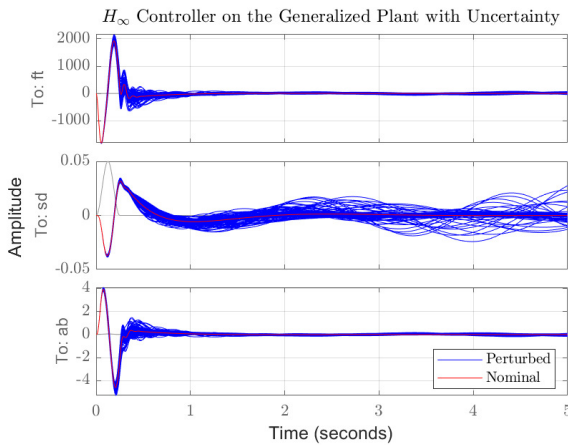


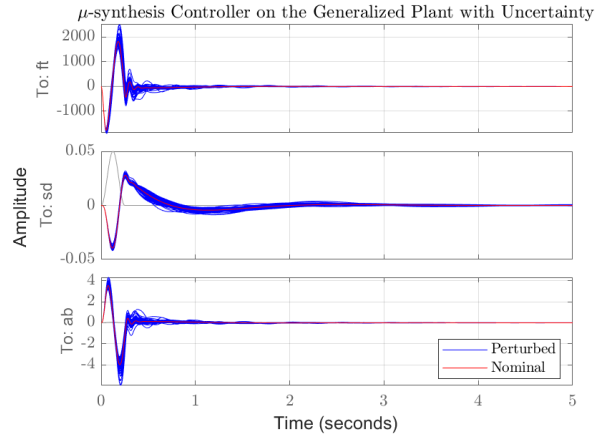
Figure 7: Outputs and Control Input for Comfort Mode

Figure 8 displays the outputs of the system with uncertainty when the comfort mode $\mathcal{H}_\infty$ controller is used. It can be observed that even though the tire force and the body acceleration approach steady state, some of the models create constant or increasing oscillations for the suspension deflection, resulting in unstable closed loop systems with the chosen controller.

Figure 8: $\mathcal{H}_\infty$ Controller with Actuator Uncertainty

On Figure 9, the system with uncertainty is under the influence of the μ -synthesis controller. While approaching zero steady state for tire force and body acceleration, the magnitude of oscillations for this system is less compared to

the $\mathcal{H}_\infty$ control. The responses for the uncertainty modes have lower settling times overall. Additionally, all of the systems appear stable for the suspension deflection. Thus, it can be concluded that the instability displayed by some of the plants controlled by the $\mathcal{H}_\infty$ can be stabilized by the μ -synthesis controller.

Figure 9: μ -synthesis Controller with Actuator Uncertainty

IV. MODEL PREDICTIVE CONTROL

For the second control architecture a model predictive control (MPC) approach is applied and examined. Model predictive control is an advanced control strategy that uses predictions of future behaviour to find the optimal control action. To do this, a constrained optimisation problem is solved at every timestep, to find the control inputs that minimises the cost function. The first control input, u_1 , is then applied to the system, the system states are updated and the process is repeated at the next timestep.

There are a number of benefits to this approach. The cost functions contains weight factors that can be tuned to reflect the importance of the individual states and the control action. A higher weight factor causes the corresponding state to converge faster or the corresponding control action to be less aggressive. The optimisation problem also allows for constraints to be set, to make sure the system and actuator limits are upheld. The resulting optimal solution guarantees the most efficient performance for these weights and constraints. However, solving the optimisation problem at every timestep can be computationally intensive, which can cause issues for systems with fast dynamics or limited computational power. As the performance relies on the accuracy of the prediction, a well defined high-fidelity discretized system model is required.

A. MPC problem formulation

The control objective is to stabilize the vehicle, thus model predictive control will be used for regulation. This means that given some initial condition, the controller will try to push the states to zero, actively rejecting any disturbance that is introduced into the system. Figure 10 shows a diagram of the system.

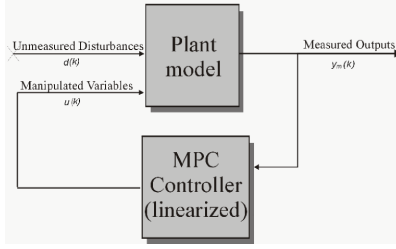


Figure 10: System diagram with MPC

First, the continuous system is discretized using zero order hold (ZOH), meaning the value of each measurement will be seen as constant until it is updated by the next measurement. The sampling time is chosen to be 0.1 seconds to ensure the discretized system is still observable: a property that is required to properly estimate the states with an observer. Also, controllability is preserved, ensuring effective, stabilizing control and enabling performance optimisation. If not all states are accessible, a state estimator has to be implemented to enable full state feedback. For the chosen method of output feedback optimisation, which directly uses the accessible system outputs, an state estimator is not necessary. This is possible because MPC aims to reduce the error of the output.

A prediction horizon N_p of 40 (time steps) is chosen, which translates to 4 seconds. This length is selected to balance computational efficiency and performance. Specifically, it has to be set to a length that ensures the optimisation solver time does not surpass the sampling time, while still providing a sufficiently large horizon for effective prediction and control performance.

Next, a Control Horizon N_c of 4 (time steps) is selected. This is done to reduce the number of optimisation variables. In the current scenario, where the control horizon is set to be less than the prediction horizon, the solver is essentially instructed to hold the control input constant beyond time step $N_c - 1$ until the prediction horizon ($N_p = 40$). Thus, the objective function will be minimised by varying only 4 variables. This ensures the optimisation problem solver is less computationally intensive, and enables faster control action.

The MPC cost function has the following form:

$$J_N(x_0, \mathbf{u}) = \sum_{k=0}^{N-1} \ell(x(k), u(k)) + V_f(x(N)) \quad (6)$$

Here, $\ell(x(k), u(k))$ holds the stage costs described by:

$$\ell(x(k), u(k)) = \frac{1}{2} (x(k)^\top Q x(k) + u(k)^\top R u(k)) \quad (7)$$

$V_f(x(N))$ holds the terminal cost, which puts additional weight on the states after the prediction horizon. This term is omitted, because of the large prediction horizon. The state costs are sufficient to reach satisfactory states before the prediction horizon is reached.

The first state cost weight affects the tire force and road holding, which is important for grip and safe travels. The objective is to keep low frequency body acceleration to a minimum, while high frequency body acceleration is allowed.

Both the second and third term on the diagonal affect body acceleration. The second term, corresponding to suspension deflection, is relatively large, because it influences the lower frequency body acceleration. The final weight affects mostly the high frequency response and is kept at zero. Although the actuator constraint keeps the control action within safe margins, a small weight R is applied to enforce efficiency. Some final adjustments can then be tuned iteratively.

$$Q = \begin{bmatrix} 4e4 & 0 & 0 \\ 0 & 5e3 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad R = 1 \quad (8)$$

In order to ensure the feasibility of the problem in a real life application, state and input constraints need to be defined. The following the physical limitations of the system are covered:

- Tire Force: $-5000 \text{ N} \leq x_1 \leq 5000 \text{ N}$
- Body Acceleration: $-5 \text{ m} \leq x_2 \leq 5 \text{ m}$
- Suspension deflection: $-0.05 \text{ m} \leq x_3 \leq 0.05 \text{ m}$

Finally, the input constraints were chosen to keep the actuator force within operating bounds.

$$-5 \text{ kN} \leq u(k) \leq 5 \text{ kN}$$

To conclude, the optimisation problem that is solved during each iteration is expressed as:

$$P_N(x_0, t) : \begin{cases} \min_{\mathbf{u}_N} & J_N(x_0, \mathbf{u}) \\ \text{s.t.} & x^+ = Ax + Bu \\ & Fx(k) \leq e \\ & |u(k)| \leq f_{max} \end{cases} \quad (9)$$

The first constraint updates the states for each time step in the prediction horizon, ensuring that the system dynamics are respected. The second and third constraints contain the system limitations and actuator limits as described previously.

B. Simulation disturbance rejection MPC vs. LQR

Now that the problem is fully defined, the model predictive control approach can be implemented and simulated in MATLAB. For the implementation the MATLAB Model Predictive Control Toolbox [5] will suffice and is chosen because of its simplicity. The full code can be found in Appendix C. Next to this, the performance of a simple Linear-Quadratic Regulator (LQR) controller with equal Q and R matrices is simulated as a benchmark to compare to the model predictive control design.

First, the system will be disturbed by a step of 5cm at 0.5s, which simulates a sudden increase in road height. The disturbance and the response of the suspension without an actuator are shown in Figure 13 in Appendix A. The disturbance responses for the system with LQR control and model predictive control is simulated. The resulting states can be seen in Figure 11.

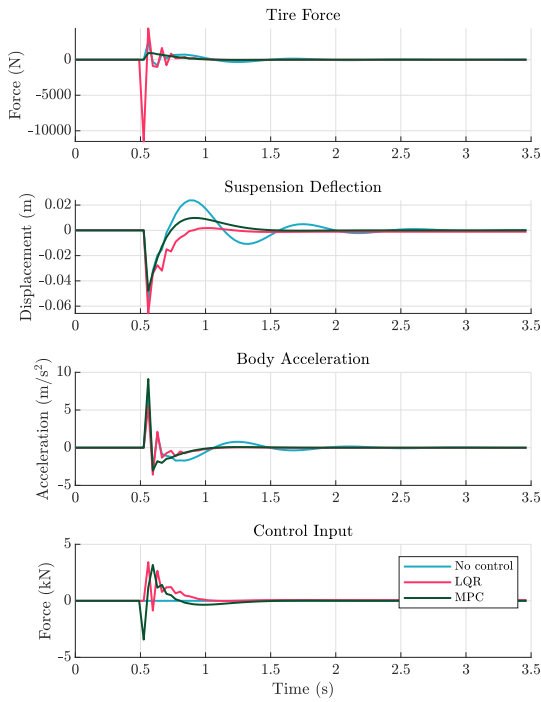


Figure 11: 5cm step LQR vs. MPC

Figure 11 shows an improvement in suspension deflection and body acceleration for both LQR and MPC compared to the uncontrolled situation in Figure 13. The tire force exhibits significant spikes for LQR, while MPC demonstrates better road holding. MPC shows a smaller yet notable improvement in suspension deflection and body acceleration compared to LQR. It is important to note that due to the relatively large time step, the simulations are prone to numerical errors; small changes in the step size can yield significantly different results. There may be an opportunity to decrease the prediction horizon, allowing for a smaller time step.

For the final simulation, the system is disturbed from 0.5 seconds by a series of three sinusoidal-shaped bumps, each 5 cm in height and lasting 1 second. This 1 Hz excitation is an uncomfortable low frequency, so a low body acceleration is desired. The system's uncontrolled response to the disturbance is displayed in Figure 14 in Appendix A. Next, the system's responses to the sinusoidal disturbance are simulated for both LQR control and model predictive control. The resulting states are shown below in Figure 12.

Figure 12 clearly shows large improvements for all responses for both LQR and MPC compared to the uncontrolled situation in Figure 14. Although the suspension deflection appears slightly more favorable for LQR, MPC shows a desirable improvement in body acceleration. In general, MPC shows results fairly similar to LQR. However, MPC still holds the potential for further tuning and has the advantage of implemented constraints, ensuring the system's limitations are taken into account when optimising the control input.

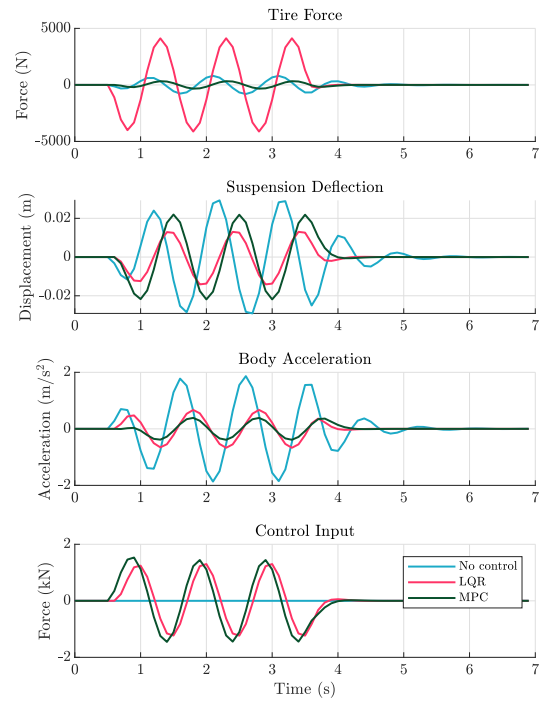


Figure 12: 3 sinusoidal bumps LQR vs. MPC

V. CONCLUSION

This report presents the design and implementation of robust $\mathcal{H}_\infty$ and MPC control approaches for active suspension. The controllers were developed and simulated on a quarter car model, which simplifies the computations while capturing the essential dynamics.

The $\mathcal{H}_\infty$ controller provides faster disturbance rejection with all three of the discussed designs (Comfort, Balanced, Handling) compared to the open-loop system. The comfort mode allows for the uneven road profile to be felt less by the passengers by slower suspension deflection and lower body acceleration while a more rapid response occurs in the handling mode. To that end, a larger control action is generated by the controller of this mode. When uncertainty is introduced to the system, stability for suspension deflection can not be achieved for some of the perturbed models with the $\mathcal{H}_\infty$ controller. By implementing a μ -synthesis design, these models are also forced to stability with a higher control action and the closed loop system achieves robust performance. Thus, even though both controllers can be tuned to provide the users with different comfort levels, μ -synthesis should be chosen to account for the mismatch between the real system and the approximated model. This controller can be beneficial for situations with considerable uncertainty in both model and conditions, like for the extreme case of off-road racing in rough weather.

The proposed MPC controller drastically reduces the effect of road disturbances on suspension deflection and body acceleration, while maintaining proper road holding. Both the amplitude and settling time are significantly decreased compared to the uncontrolled case. Notably, LQR showed

very similar results, but offers fewer tuning opportunities and lacks the benefit of optimising while explicitly handling input and output constraints. A more complex approach, without using the MPC toolbox, could be explored for more insights and control. More extensive tuning of the horizons and weights can further improve performance, and finally, a simple root mean square (RMS) of the body acceleration can be computed, to quantify the performance for easier comparison. In conclusion, the MPC controller is best suited for applications with minimal model uncertainty and high-performance requirements, such as racing on a permanent track with a well-understood and maintained vehicle.

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APPENDIX A ADDITIONAL FIGURES

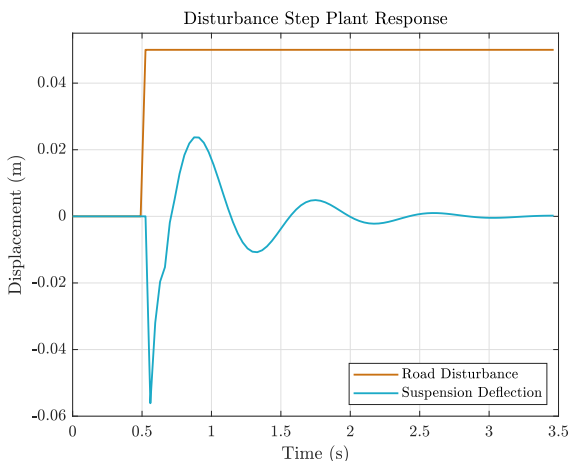


Figure 13: 5cm step uncontrolled response

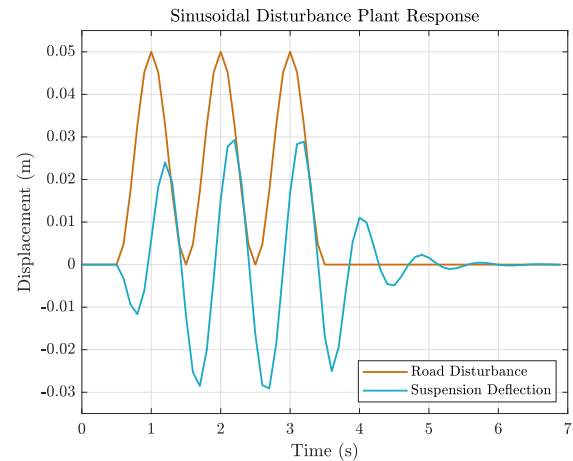


Figure 14: 3 sinusoidal bumps uncontrolled response

APPENDIX B H_INFINITY_GROUP10.M

```
% Vehicle Dynamics and Control
% Project Group 10
% H-infinity Controller for Active
% Suspension of a Quarter Car Model
% Code based on https://nl.mathworks.com/
% help/robust/gs/active-suspension-
% control-design.html

clc;
clear;
close all;
%% Create Directory for plots
if isunix
    command = 'rm -r plots';
elseif ispc
    command = 'rmdir /s /q plots';
elseif ismac
    command = 'rm -r plots';
end

status = system(command);
if (status ~= 0)
    disp("Failed to remove 'plots'
        directory. Please remove it
        manually")
end

command = 'mkdir plots';
status = system(command);
if (status ~= 0)
    disp("Failed to create 'plots'
        directory. Please create it
        manually")
end
%% Set interpreter and layout options
set(groot, 'defaulttextinterpreter', 'latex
    ');
```

```

set(groot,'
    defaultAxesTickLabelInterpreter','
    latex');
set(groot,'defaultLegendInterpreter','
    latex');
set(groot,'defaultLineLineWidth',0.8)
set(groot,'defaultAxesFontSize',11)
set(groot,'defaultAxesFontWeight',"normal
    ")

%% QC parameters
m_s = 410;

    % sprung mass, kg (Body Car)
m_u = 45;

    % unsprung mass, kg (Tire)
k_t = 230000;

    % tire vertical stiffness, N/m
k_s = 25500;

    % suspension vertical stiffness, N/m
wn_s = sqrt(k_s*k_t / (k_s + k_t)/m_s);
    % sprung mass natural
    frequency
d_s = 0.3 * (2 * m_s * wn_s);
    % damping ratio

% frequency range determination
n_points = 5000;
w = linspace(0.1,100,n_points)*2*pi;

%%
% state matrix
A = [0, 1, 0, 0;...
    %
    unsprung displacement - road
    disturbance
    -k_t / m_u, -d_s / m_u, k_s / m_u,
    d_s / m_u;... % unsprung mass
    acceleration
    0, -1, 0, 1;...
    %
    sprung displacement - unsprung
    displacement
    0, d_s / m_s, -k_s / m_s, -d_s / m_s
    ];
    % sprung mass
    acceleration
B = [0, m_s / m_u, 0, -1]';
    % control
    matrix

% Road Disturbance
G = [0, k_t / m_u, 0, 0]';

% output matrix
C = [k_t 0 0 0;...

    tire force
    0 0 1 0;...

    % suspension stroke
    A(4,:)];

    % sprung mass acceleration
Dw = [-k_t; 0; 0];
Du = [0; 0; -1];

    %
    feedthrough matrix

%%
LTI.A = A;
LTI.B = [G B]; % u1: disturbance r, u2:
    fs
LTI.C = C;
LTI.D = [Dw Du];

qcar = ss(LTI.A, LTI.B, LTI.C, LTI.D);

% States
% Tire Deflection, Tire Velocity, Body
    Travel, Body Velocity

qcar.StateName = {'Tire Deflection (m)';
    'Tire Velocity (m/s)';...
    'Body Travel (m)'; '
    Body Velocity (m/s)'
    };

qcar.InputName = {'r'; 'fs'};

% Outputs - Tire Force, Suspension
    Deflection, Body Acceleration
qcar.OutputName = {'ft'; 'sd'; 'ab'};

%%
figure;
bodemag(qcar({'ab','sd'}, 'r'), 'b', qcar({'
    ab','sd'}, 'fs'), 'r', {1 100});
legend('Road disturbance (r)', 'Actuator
    force (fs)', 'location', 'SouthWest')
title({'Gain from road dist (r) and
    actuator force (fs) ';
    'to body accel (ab) and suspension
    travel (sd)'});

grid minor
%%
%ActNom = tf(1,[1/60 1]);
ActNom = tf(1,[1/m_u 1]);

Wunc = makeweight(0.40,15,3);
unc = ultidyn('unc',[1 1], 'SampleStateDim
    ',5);
Act = ActNom*(1 + Wunc*unc);
Act.InputName = 'u';

```



```

Act.OutputName = 'fs';

rng('default');
f = figure("Name","Uncertainty");
bode(Act,'b',Act.NominalValue,'r+',
      logspace(-1,3,120));
title("Uncertainty in the Actuator Model
      for various Frequencies");
grid minor
saveas(f, "plots/Uncertain_act",'epsc');

%%
Wroad = ss(0.07); Wroad.u = 'd1';
Wroad.y = 'r';
Wact = 0.8*tf([1 50],[1 500]); Wact.u =
'u'; Wact.y = 'e1';
Wd2 = ss(0.01); Wd2.u = 'd2'; Wd2.y =
'Wd2';
Wd3 = ss(0.5); Wd3.u = 'd3'; Wd3.y =
'Wd3';

HandlingTarget = 0.04 * tf([1/8 1],[1/80
1]);
ComfortTarget = 0.4 * tf([1/0.45
1],[1/150 1]);

Targets = [HandlingTarget ; ComfortTarget
];
figure;
bodemag(qcar({'sd','ab'},'r')*Wroad,'b',
Targets,'r--',{1,1000}), grid
title('Response to road disturbance')
legend('Open-loop','Closed-loop target');
grid minor
%%
% Three design points
beta = reshape([0.01 0.5 0.99],[1 1 3]);
Wsd = beta / HandlingTarget;
Wsd.u = 'sd'; Wsd.y = 'e2';
Wab = (1-beta) / ComfortTarget;
Wab.u = 'ab'; Wab.y = 'e3';

sdmeas = sumblk('y1 = sd+Wd2');
abmeas = sumblk('y2 = ab+Wd3');
ICinputs = {'d1','d2','d3','u'};
ICoutputs = {'e1','e2','e3','y1','y2'};
qcaric = connect(qcar((2:3),:),Act,Wroad,
Wact,Wab,Wsd,Wd2,Wd3,...
sdmeas,abmeas,ICinputs,
ICoutputs);

ncont = 1; % one control signal, u
nmeas = 2; % two measurement signals, sd
and ab
K = ss(zeros(ncont,nmeas,3));
gamma = zeros(3,1);
for i=1:3
    [K(:,:,i),~,gamma(i)] = hinfsyn(qcaric
    (:,:,i),nmeas,ncont);
end

gamma;
%%
% Closed-loop models
K.u = {'sd','ab'}; K.y = 'u';
CL = connect(qcar,Act.Nominal,K,'r',{'ft'
'sd';'ab'});

f = figure("Name","Bode of 3 Modes");
bodemag(qcar(:, 'r'),'b', CL(:,:,1),'r-.',
...
CL(:,:,2),'m-.', CL(:,:,3),'k-.',
{1,140}), grid
legend('Open-loop','Comfort','Balanced','
Handling','location','SouthEast')
title({'Tire Force, suspension deflection
, body acceleration due to road'})
grid minor

saveas(f, 'plots/bode_hinf_3modes','epsc'
);
%%
% Road disturbance
t = 0:0.0025:5;
roaddist = zeros(size(t));
roaddist(1:101) = 0.025*(1-cos(8*pi*t
(1:101)));

% Closed-loop model
%SIMK = connect(qcar,Act.Nominal,K,'r',{'
sd';'ft';'ab';'fs'});
SIMK = connect(qcar,Act.Nominal,K,'r',{'
ft';'sd';'ab';'fs'});

% Simulate
p1 = lsim(qcar(:,1),roaddist,t);
y1 = lsim(SIMK(1:4,1,1),roaddist,t);
y2 = lsim(SIMK(1:4,1,2),roaddist,t);
y3 = lsim(SIMK(1:4,1,3),roaddist,t);

%% Comfort Controller
f = figure("Name","Comfort Controller");
% Plot results
t2 = tiledlayout(2,2);
sgtitle("Comfort Mode","Interpreter","
Latex")
nexttile
plot(t,p1(:,1),'b',t,y1(:,1),'r--'); %t,
y2(:,3),'m.',t,y3(:,3),'k.',
title('Tire Force'), ylabel('N');
xlim([0 4]);
legend('Open-loop', '$H_{\infty}$', '
Location','northeast');
grid minor

```

```

nexttile
plot(t,roaddist,'g',t,p1(:,2),'b',t,y1
(:,2),'r--'); %t,y2(:,1),'m.',t,y3
(:,1),'k.',
title('Suspension Deflection'), ylabel('$s_d$ (m)$');
legend('Disturbance', 'Open-loop', '$H_{\infty}$', 'Location', 'northeast');
xlim([0 4]);
grid minor

nexttile
plot(t,p1(:,3),'b',t,y1(:,3),'r--'); %t,
y2(:,3),'m.',t,y3(:,3),'k.',
title('Body acceleration'), ylabel('$a_b$ (m/s^2)$');
legend('Open-loop', '$H_{\infty}$', 'Location', 'northeast');
xlim([0 4]);
grid minor

nexttile
plot(t,y1(:,4),'r'); %t,y2(:,3),'m.',t,y3
(:,3),'k.',
title('Control force'), xlabel('Time (s)')
, ylabel('$f_s$ (kN)$');
xlim([0 4]);
grid minor

saveas(f, 'plots/comfort_controller','
eps');

%% Balanced Controller
f = figure("Name","Balanced Controller");
% Plot results
t3 = tiledlayout(2,2);
sgtitle("Balanced Mode","Interpreter","
Latex")
nexttile
plot(t,p1(:,1),'b',t,y2(:,1),'r--');
title('Tire Force'), ylabel('N');
xlim([0 4]);
legend('Open-loop', '$H_{\infty}$', 'Location', 'northeast');
grid minor

nexttile
plot(t,roaddist,'g',t,p1(:,2),'b',t,y2
(:,2),'r--');
title('Suspension Deflection'), ylabel('$s_d$ (m)$');
legend('Disturbance', 'Open-loop', '$H_{\infty}$', 'Location', 'northeast');
xlim([0 4]);
grid minor

nexttile
plot(t,p1(:,3),'b',t,y3(:,3),'r--');
title('Body acceleration'), ylabel('$a_b$ (m/s^2)$');
legend('Open-loop', '$H_{\infty}$', 'Location', 'northeast');
xlim([0 4]);
grid minor

nexttile
plot(t,y3(:,4),'r');
title('Control force'), xlabel('Time (s)')
, ylabel('$f_s$ (kN)$');
xlim([0 4]);
grid minor

title('Body acceleration'), ylabel('$a_b$ (m/s^2)$');
legend('Open-loop', '$H_{\infty}$', 'Location', 'northeast');
xlim([0 4]);
grid minor

nexttile
plot(t,y2(:,4),'r');
title('Control force'), xlabel('Time (s)')
, ylabel('$f_s$ (kN)$');
xlim([0 4]);
grid minor

saveas(f, 'plots/balanced_controller','
eps');

%% Handling Controller
f = figure("Name","Handling Controller");
% Plot results
t4 = tiledlayout(2,2);
sgtitle("Handling Mode","Interpreter","
Latex")
nexttile
plot(t,p1(:,1),'b',t,y3(:,1),'r--');
title('Tire Force'), ylabel('N');
xlim([0 4]);
legend('Open-loop', '$H_{\infty}$', 'Location', 'northeast');
grid minor

nexttile
plot(t,roaddist,'g',t,p1(:,2),'b',t,y3
(:,2),'r--');
title('Suspension Deflection'), ylabel('$s_d$ (m)$');
legend('Disturbance', 'Open-loop', '$H_{\infty}$', 'Location', 'northeast');
xlim([0 4]);
grid minor

nexttile
plot(t,p1(:,3),'b',t,y3(:,3),'r--');
title('Body acceleration'), ylabel('$a_b$ (m/s^2)$');
legend('Open-loop', '$H_{\infty}$', 'Location', 'northeast');
xlim([0 4]);
grid minor

nexttile
plot(t,y3(:,4),'r');
title('Control force'), xlabel('Time (s)')
, ylabel('$f_s$ (kN)$');
xlim([0 4]);
grid minor

```

```

saveas(f, 'plots/handling_controller', '
    epsc');

%%

f = figure("Name", "Comparison of All
    Modes", "Position", [500 200 500 650]);
% Plot results
t5 = tiledlayout(4,1);
sgtitle("Comparison", "Interpreter", "Latex
    ")
nexttile
hold on
plot(t, y1(:,1), 'r', t, y2(:,1), 'b')
plot(t, y3(:,1), 'Color', "#4DBEEE")
plot(t, p1(:,1), 'k--')
title('Tire Force'), ylabel('N');
xlim([0 3.5]);
legend('Comfort', 'Balanced', 'Handling',
    'Open-loop', 'Location', 'northeast');
grid minor

nexttile
hold on
plot(t, roaddist, 'g')
plot(t, y1(:,2), 'r', t, y2(:,2), 'b')
plot(t, y3(:,2), 'Color', "#4DBEEE")
plot(t, p1(:,2), 'k--')
title('Suspension Deflection'), ylabel('$
    s_d (m)$');
legend('Disturbance', 'Comfort', 'Balanced
    ', 'Handling', 'Open-loop', 'Location',
    'northeast');
xlim([0 3.5]);
grid minor

nexttile
hold on
plot(t, y1(:,3), 'r', t, y2(:,3), 'b')
plot(t, y3(:,3), 'Color', "#4DBEEE")
plot(t, p1(:,3), 'k--')
title('Body Acceleration'), ylabel('$a_b
    (m/s^2)$');
legend('Comfort', 'Balanced', 'Handling',
    'Open-loop', 'Location', 'northeast');
xlim([0 3.5]);
grid minor

nexttile
hold on
plot(t, y1(:,4), 'r', t, y2(:,4), 'b')
plot(t, y3(:,4), 'Color', "#4DBEEE");
title('Control Input'), xlabel('Time (s)')
    , ylabel('$f_s (kN)$');
legend('Comfort', 'Balanced', 'Handling',
    'Location', 'northeast');
xlim([0 3.5]);

```

```

grid minor

saveas(f, 'plots/
    mode_comparison_controller', 'epsc');
%%

[Krob, rpMU] = musyn(qcaric(:, :, 1), nmeas,
    ncont);
%%
% Closed-loop model (nominal)
Krob.u = {'sd', 'ab'};
Krob.y = 'u';
SIMKrob = connect(qcar, Act.Nominal, Krob, '
    r', {'ft'; 'sd'; 'ab'; 'fs'});

% Simulate
p1 = lsim(qcar(:, 1), roaddist, t);
y1_r = lsim(SIMKrob(1:4, 1), roaddist, t);

% Plot results
f = figure('Position', [200 200 500 650]);
sgtitle("Comfort Mode Comparison")
subplot(411)
plot(t, y1(:,1), 'b', t, y1_r(:,1), 'r', t, p1
    (:,1), 'k--')
title('Tire Force'), ylabel('$f_t (N)$')
legend('$H_{\infty}$', '$\mu$-syn', 'Open-
    loop', 'Location', 'northeast');
xlim([0,3])
grid minor
subplot(412)
plot(t, roaddist, 'g', t, y1(:,2), 'b', t, y1_r
    (:,2), 'r', t, p1(:,2), 'k--')
title('Suspension Deflection'), xlabel('
    Time (s)'), ylabel('$s_d (m)$')
legend('Disturbance', '$H_{\infty}$', '$\mu$
    -syn', 'Open-loop', 'Location', '
    northeast');
xlim([0,3])
grid minor
subplot(413)
plot(t, y1(:,3), 'b', t, y1_r(:,3), 'r', t, p1
    (:,3), 'k--')
title('Body Acceleration'), ylabel('$a_b
    (m/s^2)$')
legend('$H_{\infty}$', '$\mu$-syn', 'Open-
    loop', 'Location', 'northeast');
xlim([0,3])
grid minor
subplot(414)
plot(t, y1(:,4), 'b', t, y1_r(:,4), 'r')
title('Control Input'), xlabel('Time (s)')
    , ylabel('$f_s (kN)$')
legend('$H_{\infty}$', '$\mu$-syn', '
    Location', 'northeast');
xlim([0,3])
grid minor

```

```

saveas(f, "plots/Robust_Controller", '
    epsc');

%%
rng('default'), nsamp = 100;

% Uncertain closed-loop model with
% comfort H-infinity controller
CLU = connect(qcar,Act,K(:, :, 1), 'r', {'ft'
    , 'sd', 'ab'});
f = figure;
lsim(usample(CLU, nsamp), 'b', CLU.Nominal, '
    r', roaddist, t)
title('$H_{\infty}$ Controller on the
    Generalized Plant with Uncertainty', '
    Interpreter', 'latex')
legend('Perturbed', 'Nominal', 'location', '
    SouthEast')
grid minor
saveas(f, "plots/Hinf_Controller_Comfort
    ", 'epsc');

%%

% Uncertain closed-loop model with
% comfort robust controller
CLU = connect(qcar,Act,Krob,'r',{'ft','sd'
    , 'ab'});
f = figure;
lsim(usample(CLU, nsamp), 'b', CLU.Nominal, '
    r', roaddist, t)
title('$\mu$-synthesis Controller on the
    Generalized Plant with Uncertainty', '
    Interpreter', 'latex')
legend('Perturbed', 'Nominal', 'location', '
    SouthEast')
grid minor
saveas(f, "plots/
    Robust_Controller_Comfort", 'epsc');

```

APPENDIX C

MPC_CONTROLLER_GROUP10.M

```

%% Vehicle Dynamics and Control
% Project Group 10
% MPC Controller for Active Suspension of
% a Quarter Car Model

clc;
clear;
close all;

%%
if isunix
    command = 'rm -r plots_mpc';
elseif ispc
    command = 'rmdir /s /q plots_mpc';
elseif ismac
    command = 'rm -r plots_mpc';

```

```

end

status = system(command);
if (status ~= 0)
    disp("Failed to remove 'plots_mpc'
        directory. Please remove it
        manually")
end

command = 'mkdir plots_mpc';
status = system(command);
if (status ~= 0)
    disp("Failed to create 'plots'
        directory. Please create it
        manually")
end

% Set interpreter and layout options
set(groot, 'defaulttextinterpreter', 'latex
    ');
set(groot, '
    defaultAxesTickLabelInterpreter', '
    latex');
set(groot, 'defaultLegendInterpreter', '
    latex');
set(groot, 'defaultLineLineWidth', 0.8)
set(groot, 'defaultAxesFontSize', 11)
set(groot, 'defaultAxesFontWeight', "normal
    ")

%%
% QC parameters
m_s = 410;

% sprung mass, kg
m_u = 45;

% unsprung mass, kg
k_t = 230000;

% tire vertical stiffness, N/m
k_s = 25500;

% suspension vertical stiffness, N/m
wn_s = sqrt(k_s*k_t / (k_s + k_t)/m_s);
% sprung mass natural
frequency
d_s = 0.3 * (2 * m_s * wn_s);
% damping ratio

% frequency range determination
n_points = 5000;
w = linspace(0.1, 100, n_points)*2*pi;

%%
% passive QC
% state matrix
A = [0, 1, 0, 0; ...

```

```

                                %
unsprung displacement - road
disturbance
-k_t / m_u, -d_s / m_u, k_s / m_u,
  d_s / m_u;... % unsprung mass
  acceleration
0, -1, 0, 1;...

    sprung displacement - unsprung
    displacement
0, d_s / m_s, -k_s / m_s, -d_s / m_s
];          % sprung mass
  acceleration
B = [0, m_s / m_u, 0, -1]';

                                % control
                                matrix

% Road Disturbance
G = [0, k_t / m_u, 0, 0]';

% output matrix
C = [k_t 0 0 0;...

                                %
    tire force
    0 0 1 0;...

    % suspension stroke
    A(4,:)];

    % sprung mass acceleration
Dw = [-k_t; 0; 0];
Du = [0; 0; -1];

                                %
    feedthrough matrix

%% Augment the Matrix
lti.A = A;
lti.B = [B G];
lti.C = C;
lti.D = [Du Dw];

plant = ss(lti.A,lti.B,lti.C,lti.D);

trans_fun = tf(plant);

plant = tf(trans_fun.Numerator, trans_fun
.Denominator,'OutputDelay', 0.0000001)
;

Ts = 0.1; % (Numerical issues) 0.035 for
step results, 0.1 for sinusoidal
results
Tstop = 7; % SIMULATION TIME [s] (7s for
Sinusoidal of 3s, 3.5s for step)
sim_time = round(Tstop/Ts);

% Discretizes plant, obsolete 'cause mpc
(.) does so, required for quadprog

                                % plant = c2d(plant,Ts);

%% Road disturbance generation
% Define the frequency of the road
disturbance.
f = 1; % Hz [4 to compare with Robust
controller]

% Disturbance_start = 0.5;      % Play around
with [s]
Disturbance_duration = 3;      % Play around
with [s]
Disturbance_end = Disturbance_start+(1/f)
*floor(Disturbance_duration/(1/f));

% Create a time vector from 0 to Tstop.
% with increments of Ts.
T = (0:Ts:Tstop - Ts)';

% Define the road disturbance profile.
v = 0.025*(1-cos(2*pi*f*(T-
Disturbance_start)));          % was 0.05;
0.025 to compare with H-\infty

% Oscillating disturbance Start and Stop
times
% v(T < 2*1/f) = 0;
% v(T > 3*1/f) = 0;
v(T < Disturbance_start) = 0;
v(T > Disturbance_end) = 0;

% Step disturbance (COMMENT FOR SINUSOID;
uncomment for step)
% v = zeros(size(T));
% v(T>Disturbance_start) = 0.05;

%%
% T = (0:Ts:(sim_time*Ts - Ts))';
Unn = zeros(sim_time,2);
Unn(:,2) = v; %hx(1:sim_time);

[Yuc,Tuc,Xuc] = lsim(plant,Unn,T);

f = figure;
title('Response of Uncontrolled
Suspension to Road Disturbance');
plot(T, v, 'DisplayName', 'Road
Disturbance','color','#c97112',"
LineWidth",1.25); hold on;
% plot(Tuc, Yuc(:,1)*1e-3, 'DisplayName',
'Body acceleratoin','color','#1caac7
',"LineWidth",1.25); % Scaled body
acceleration response
plot(Tuc, Yuc(:,2), 'DisplayName', '
Suspension Deflection','color','#1
caac7',"LineWidth",1.25); % Comment
out for only Step OR Sinusoid
xlim([0,Tstop])

```



```

% ylim([-0.01,0.06])      % Uncomment for
    Sinusoid_Disturbance w/ Ts = 0.01 OR
    Step_Disturbance with Ts = 0.07
% ylim([-0.06,0.055])    % Uncomment for
    Step_Plant_Response with Ts = 0.07
ylim([-0.035,0.055])     % Uncomment for
    Sinusoid_Plant_Response w/ Ts = 0.01
    OR

hold off;
grid on;
legend('location','southeast');

% title('Sinusoidal Disturbance')
    % Uncomment for
    Sinusoid_Disturbance
title('Sinusoidal Disturbance Plant
Response') % Uncomment for
    Sinusoid_Plant_Response
% title('Disturbance Step')
    % Uncomment for
    Step_Disturbance
% title('Disturbance Step Plant Response
') % Uncomment for
    Step_Plant_Response

ylabel('Displacement (m)');
xlabel('Time (s)');
saveas(f, 'plots_mpc/disturbance', 'epsc'
);

%% LQR

% Weights selection for use in
    performance index
r1 = 4e+4;      % road holding weight
r2 = 5e+3;      % comfort weight
r3 = 0;         % control effort weight
Rxx = A(4,:)'*A(4,:) + diag([r1 0 r2 0]);
Rxu = -A(4,:);
Ruu = 1 + r3;

% LQR optimal gain
[Kr,~] = lqr(A,B,Rxx,Ruu,Rxu);

% Closed Loop System
Ac = (A - B * Kr);
Cc = (C - Du * Kr);

plant_lqr = ss(Ac, G, Cc, Dw);

plant_lqr = c2d(plant_lqr, Ts);

[y_lqr, time, stuff] = lsim(plant_lqr, v,
    T);

Ulqr = -(Kr*stuff)';

%%
plant = setmpcsignals(plant,'MD',2);

p = 40;      % Prediction horizon
m = 4;      % Control horizon
mpcobj = mpc(plant,Ts,p,m);

% Tire force [N]
mpcobj.OutputVariables(1).Min = -5000;
mpcobj.OutputVariables(1).Max = 5000;

% Suspension deflection [m]
mpcobj.OutputVariables(2).Min = -0.05;
mpcobj.OutputVariables(2).Max = 0.05;

% Body acceleration [m/s^2]
mpcobj.OutputVariables(3).Min = -5;
mpcobj.OutputVariables(3).Max = 5;

% Control action stays within actuator
    limits
mpcobj.ManipulatedVariables(1).Min =
    -5000;
mpcobj.ManipulatedVariables(1).Max =
    5000;

% Scale factors
mpcobj.OutputVariables(2).ScaleFactor = 1
    e-5;
mpcobj.OutputVariables(3).ScaleFactor = 1
    e-3;

% WEIGHTS
% Control action weight (same as LQR)
mpcobj.Weights.ManipulatedVariables = Ruu
    ;
mpcobj.Weights.ManipulatedVariablesRate =
    1;
% State weights (same as LQR) [Tire force
    , suspension defl., body acc.]
mpcobj.Weights.OutputVariables = [r1 r2
    0];

review(mpcobj);

%%
r = zeros(sim_time,3);
v = reshape(v,length(v),1);

params = mpcsimopt();
params.MDLookAhead = 'off';

mpcobj.Optimizer;
[Yas, Tas, Uas, XPas, XCas] = sim(mpcobj,
    sim_time,r,v,params);

f = figure('Position',[100 100 500 650]);
t1 = tiledlayout(4,1);

```

```

nexttile;
hold on
plot(Tuc, Yuc(:,1), 'color', '#1caac7', "
      LineWidth",1.25);
plot(time, y_lqr(:,1), 'color', '#ff3366', "
      LineWidth",1.25);
plot(Tas, Yas(:,1), 'color', '#09512d', "
      LineWidth",1.25);
xlim([0,Tstop])
hold off;
% grid minor;
grid on;
title("Tire Force");
ylabel("Force (N) ")
% xlabel("Time (s) ")

% grid minor;
grid on
title("Control Input");
% legend("No control","MPC", "location","
      northeast","FontSize",7);
legend("No control","LQR","MPC", "
      location","northeast","FontSize",7);
% legend("No control","LQR","MPC", "
      location","southeast","FontSize",7);
ylabel("Force (kN) ")
xlabel("Time (s) ")

saveas(f, 'plots_mpc/MPC_outputs', 'epsc'
      );

nexttile;
hold on
plot(Tuc, Yuc(:,2), 'color', '#1caac7', "
      LineWidth",1.25);
plot(time, y_lqr(:,2), 'color', '#ff3366', "
      LineWidth",1.25);
plot(Tas, Yas(:,2), 'color', '#09512d', "
      LineWidth",1.25);
xlim([0,Tstop])
hold off;
% grid minor;
grid on
title("Suspension Deflection");
ylabel("Displacement (m) ")
% xlabel("Time (s) ")

nexttile;
hold on
plot(Tuc, Yuc(:,3), 'color', '#1caac7', "
      LineWidth",1.25);
plot(time, y_lqr(:,3), 'color', '#ff3366', "
      LineWidth",1.25);
plot(Tas, Yas(:,3), 'color', '#09512d', "
      LineWidth",1.25);
xlim([0,Tstop])
hold off;
% grid minor;
grid on
title("Body Acceleration");
ylabel("Acceleration (m/s^2) ")
% xlabel("Time (s) ")

nexttile;
hold on
plot(Tuc, Tuc*0, 'color', '#1caac7', "
      LineWidth",1.25);
plot(time, Ulqr, 'color', '#ff3366', "
      LineWidth", 1.25);
plot(Tas, Uas, 'color', '#09512d', "
      LineWidth",1.25);
xlim([0,Tstop])
hold off;

```